FractionalMaxPool2d#

https://pytorch.org/docs/stable/generated/torch.nn.FractionalMaxPool2d.html

Description:

Applies a 2D fractional max pooling over an input signal composed of several input planes. Fractional MaxPooling is described in detail in the paper Fractional MaxPooling by Ben Graham The max-pooling operation is applied in kH×kWkH \times kWkH×kW regions by a stochastic step size determined by the target output size. The number of output features is equal to the number of input planes. Exactly one of output_size or output_ratio must be defined.

Function Signature

```
class torch.nn.FractionalMaxPool2d(kernel_size,
output_size=None, output_ratio=None, return_indices=False,
_random_samples=None)[source]#

Parameters

Shape:
```

Code Examples

```
>>> # pool of square window of size=3, and target output size
13x12
>>> m = nn.FractionalMaxPool2d(3, output_size=(13, 12))
>>> # pool of square window and target output size being half of
input image size
>>> m = nn.FractionalMaxPool2d(3, output_ratio=(0.5, 0.5))
>>> input = torch.randn(20, 16, 50, 32)
>>> output = m(input)
```

Notes & Warnings

NOTE Note Exactly one of output_size or output_ratio must be defined.

Detailed Documentation

Documentation

Applies a 2D fractional max pooling over an input signal composed of several input planes.

Fractional MaxPooling is described in detail in the paper Fractional MaxPooling by Ben Graham

The max-pooling operation is applied in kH×kWkH \times kWkH×kW regions by a stochastic step size determined by the target output size. The number of output features is equal to the number of input planes.

Exactly one of output_size or output_ratio must be defined.

kernel_size (Union[int, tuple[int, int]]) – the size of the window to take a max over. Can be a single number k (for a square kernel of k x k) or a tuple (kh, kw)

output_size (Union[int, tuple[int, int]]) – the target output size of the image of the form oH x oW. Can be a tuple (oH, oW) or a single number oH for a square image oH x oH. Note that we must have kH+oH−1<=HinkH + oH - 1 <= H_{in}kH+oH−1<=Hin and kW+oW−1<=WinkW + oW - 1 <= W_{in}kW+oW−1<=Win

output_ratio (Union[float, tuple[float, float]]) – If one wants to have an output size as a ratio of the input size, this option can be given. This has to be a number or tuple in the range (0, 1). Note that we must have kH+(output_ratio_H∗Hin)−1<=HinkH + (output_ratio_H * H_{in}) - 1 <= H_{in}kH+(output_ratio_H∗Hin)−1<=Hin and kW+(output_ratio_W∗Win)−1<=WinkW + (output_ratio_W * W_{in}) - 1 <= W_{in}kW+(output_ratio_W∗Win)−1<=Win

return_indices (bool) – if True, will return the indices along with the outputs. Useful to pass to nn.MaxUnpool2d(). Default: False

Input: (N,C,Hin,Win)(N, C, H_{in}, W_{in})(N,C,Hin,Win) or (C,Hin,Win)(C, H_{in}, W_{in})(C,Hin,Win).

Output: (N,C,Hout,Wout)(N, C, H_{out}, W_{out})(N,C,Hout,Wout) or (C,Hout,Wout)(C, H_{out}, W_{out})(C,Hout,Wout), where (Hout,Wout)=output_size(H_{out}, W_{out})=\text{output_size}(Hout,Wout)=output_size or (Hout,Wout)=output_ratio×(Hin,Win)(H_{out}, W_{out})=\text{output_ratio} \times (H_{in}, W_{in})(Hout,Wout)=output_ratio×(Hin,Win).